

Visualizing Undirected Graph That's Too Large for GraphViz?

I need advice for rendering an undirected graph with 178,000 nodes and 500,000 edges. I've tried Neato, Tulip, and Cytoscape. Neato doesn't even come remotely close, and Tulip and Cytoscape claim they can handle it but don't seem to be able to. (Tulip does nothing and Cytoscape claims to be working, and then just stops.)

I'd just like a vector format file (ps or pdf) with a remotely reasonable layout of the nodes.

graph-theorygraphvizgraph-layout

edited Mar 23 at 18:44

Dominique Fortin

1,202614

asked Oct 26 '08 at 22:31

Gabe

50

Draw a small square and color it all black. :-) Sorry, I couldn't resist. – tvanfossion Oct 26 '08 at 23:41

1

What kind of data does this graph represent? Maybe, you can simplify it automatically? It's just my guess: I have no info on the data represented, so it's hard to guess. Anyway, so many nodes and edges will not be very expressive on one piece of paper... – avp Oct 27 '08 at 9:07

1

Just how big a PDF are you expecting - something tiled over several sheets of A3? – Andy Dent Jan 27 '09 at 13:18

1

@ Andy Dent, or several hundred... – Tom Neyland Aug 5 '09 at 19:48

You probably need to increase cytoscape's heap: cytoscape.wodaklab.org/wiki/... – Ron Mar 22 '11 at 3:15

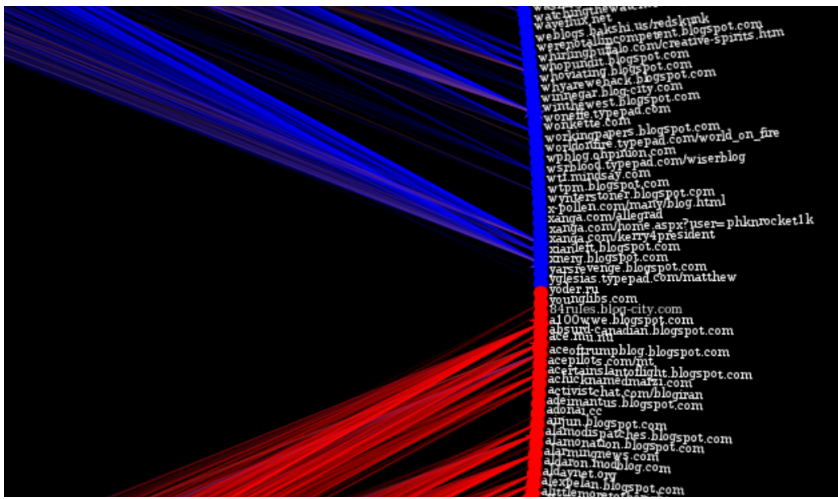
18 Answers

I've had good results using the [graph-tool](#) library in python. The below graph has 1,490 nodes and 19,090 edges - it took around 5min to render on my laptop.

The graph data comes from the political blogging network described by Adamic and Glance in ["The political blogosphere and the 2004 US Election"](#) pdf link [here](#). If you zoom in you can see the blog urls for each node.

https://stackoverflow.com/questions/238724/visualizing-undirected-graph-thats-too-large-for-graphviz

1/5



Here's the code I used to draw it (blog <http://ryancompton.net/2014/10/22/stochastic-block-model-based-edge-bundles-in-graph-tool/>):

```
import graph_tool.all as gt
import math

g = gt.collection.data["polblogs"] #
http://www2.scedu.unibo.it/roversi/SocioNet/AdamicGlanceBlogWWW.pdf
print(g.num_vertices(), g.num_edges())

#reduce to only connected nodes
g = gt.GraphView(g, vfilter=lambda v: (v.out_degree() > 0) and (v.in_degree() > 0))
g.purge_vertices()

print(g.num_vertices(), g.num_edges())

#use 1->Republican, 2->Democrat
red_blue_map = {1:(1,0,0,1), 2:(0,0,1,1)}
plot_color = g.new_vertex_property('vector<double>')
g.vertex_properties['plot_color'] = plot_color
for v in g.vertices():
    plot_color[v] = red_blue_map[g.vertex_properties['value'][v]]

#edge colors
alpha=0.15
edge_color = g.new_edge_property('vector<double>')
g.edge_properties['edge_color'] = edge_color
for e in g.edges():
    if plot_color[e.source()] != plot_color[e.target()]:
        if plot_color[e.source()] == (0,0,1,1):
            #orange on dem -> rep
            edge_color[e] = (255.0/255.0, 102/255.0, 0/255.0, alpha)
        else:
            edge_color[e] = (102.0/255.0, 51/255.0, 153/255.0, alpha)
    #red on rep-rep edges
    elif plot_color[e.source()] == (1,0,0,1):
        edge_color[e] = (1,0,0, alpha)
    #blue on dem-dem edges
    else:
        edge_color[e] = (0,0,1, alpha)

state = gt.minimize_nested_blockmodel_dl(g, deg_corr=True)
bstack = state.get_bstack()
t = gt.get_hierarchy_tree(bstack)[0]
tpos = pos = gt.radial_tree_layout(t, t.vertex(t.num_vertices() - 1), weighted=True)
cts = gt.get_hierarchy_control_points(g, t, tpos)
pos = g.own_property(tpos)
b = bstack[0].vp["b"]

#labels
text_rot = g.new_vertex_property('double')
g.vertex_properties['text_rot'] = text_rot
for v in g.vertices():
    if pos[v][0] > 0:
        text_rot[v] = math.atan(pos[v][1]/pos[v][0])
    else:
        text_rot[v] = math.pi + math.atan(pos[v][1]/pos[v][0])

gt.graph_draw(g, pos=pos, vertex_fill_color=g.vertex_properties['plot_color'],
              vertex_color=g.vertex_properties['plot_color'],
              edge_control_points=cts,
              vertex_size=10,
              vertex_text=g.vertex_properties['label'],
              vertex_text_rotation=g.vertex_properties['text_rot'],
              vertex_text_position=1,
              vertex_font_size=9,
              edge_color=g.edge_properties['edge_color'],
              vertex_anchor=0,
              bg_color=[0,0,0,1],
              output_size=[4024,4024],
              output='polblogs_blockmodel.png')
```

edited Mar 14 '15 at 20:00

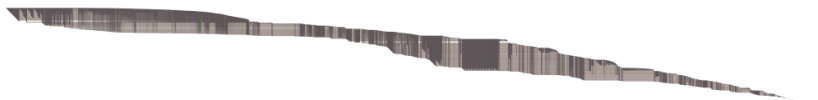
answered Oct 7 '14 at 21:46



dranzo

2,145 3 21 31

BioFabric (www.BioFabric.org) is another tool for visualizing large graphs. It should be able to handle the network described (178,000 nodes and 500,000 edges) OK, though the initial layout may take awhile. The network show here (from the Stanford Large Network Dataset Collection) is the Stanford Web Network, which has 281,903 nodes and 2,312,497 edges:



BioFabric's scalability is due to the fact that it represents nodes not as points, but as horizontal lines. The edges are then shown as vertical lines. For some intuition about how this works, there is the [Super-Quick BioFabric Demo](#), which is a small network that is animated using D3.

The primary application is written in Java. At the moment, it can only export PNG images, not PDFs. There is a PDF export option from [RBioFabric](#), though that is a very simple implementation that cannot handle really large networks yet.

Full disclosure: BioFabric is a tool that I wrote.

answered Jan 30 '15 at 5:30



wjrl
171 4

Graphviz itself provides a solution for rendering large graphs.

Namely, Graphviz includes `sfdp`, a multiscale version of `fdp` (also in graphviz, similar to `neato`) for the layout of large undirected graphs which has been useful for drawing large graphs (70k nodes, 500k edges) in my project.

You can find documentation for this software on the graphviz web site itself at <http://www.graphviz.org/>

More information, a paper describing the underlying techniques and examples can be found here: http://yifanhu.net/PUB/graph_draw_small.pdf

edited Mar 31 '14 at 19:21



Aaron Hall ♦
98.7k 29 211 196

answered Feb 23 '11 at 16:07



Anthony Liekens
707 8 22

This was the simplest solution to networkx/graphviz crashing, thanks! – [styts](#) Aug 1 '12 at 8:50

1 The link to the paper is now broken - could you include the title of the paper so that other's can trace it in the future? – [JustinJDavies](#) May 29 '13 at 8:12

I believe this is the new location for the paper:

www2.research.att.com/~yifanhu/PUB/graph_draw_small.pdf "Efficient and high quality force-directed graph drawing" by Yifan Hu. More info can be found here:

www2.research.att.com/~yifanhu/SOFTWARE/SFDP/index.html – [Anthony Liekens](#) Jun 5 '13 at 13:46

I suggest that you first do some preprocessing of the data, for example collapsing nodes to clusters and then visualizing the clusters. Collapsing will reduce the number of nodes and makes it easier for algorithms such as Kamada-Kawai or Fruchterman-Reingold to render the resulting graph.

If you really need to visualize 500,000 nodes then can you consider using a simple circular layout. This will be easy to render without the issues that force-based algorithms have. Take a look at Circos: <http://mkweb.bcgsc.ca/circos/>

Circos is graph visualization developed by bio-informatics people which is tailored to visualize genomes and other extremely large and complex data-sets.

It's a PERL based package, I hope that's not problematic.

edited Mar 31 '14 at 19:12



Aaron Hall ♦
98.7k 29 211 196

answered Jun 2 '09 at 1:36



DrDee
2,177 4 21 35

[Large Graph Layout \(LGL\)](#) project helped me a lot with a similar problem. It handles layout and have a small java app to draw produced layouts in 2D. No vector output out of the box so you'll have to draw the graph yourself (given the node coordinates produced by LGL)

edited Jul 14 '13 at 3:30



samoz
23.9k 44 119 174

answered Nov 3 '09 at 21:54



Nikita Nemkin
11 1

Link is broken as of 7/2013 – [samo](#) Jul 14 '13 at 3:19

First, I would like to second aliekens' suggestion to try `sfdp`. It is the large scale version of `Neato`.

As OJW suggests you could also just plot the nodes in R2. Your edges actually supply what he calls a "natural ordering." In particular you can plot the components of the second and third eigenvectors of the normalized graph Laplacian. This is the matrix L in [this wikipedia page about spectral clustering](#). You should be able to write down this matrix without understanding the linear algebra behind it. Then, you have reduced your problem to approximately computing the first few eigenvectors of a large sparse matrix. This is traditionally done by iterative methods and is implemented in standard linear algebra packages. This method should scale up to very large graphs.

answered Jun 19 '12 at 0:46



[MRocklin](#)

15.9k 6 46 92

Try [Gephi](#), it has a new layout plugin called [OpenOrd](#) that scales to millions of nodes.

answered Feb 20 '11 at 23:35



[Ollie Glass](#)

7,108 14 48 84

You can also try [NAViGaTOR](#) (disclosure: I'm one of the developers for that software). We've successfully visualized graphs with as many as 1.7 million edges with it. Although such large networks are hard to manipulate (the user interface will get laggy). However, it does use OpenGL for the visualization so some of the overhead is transferred to the graphics card.

Also note that you'll have to crank up the memory settings in the File->Preferences dialog box before you can successfully open a network that big.

Finally, as most of the other responses point out, you are better off re-organizing your data into something smaller and more meaningful.

answered Apr 29 '10 at 21:12



[Alinium](#)

821 1 11 22

I expect edge clustering (http://www.visualcomplexity.com/vc/project_details.cfm?id=679&index=679&domain=) would help. This technique bundles related edges together, reducing the visual complexity of the graph. You may have to implement the algorithm yourself though.

answered Aug 27 '09 at 10:15



[Ollie Glass](#)

7,108 14 48 84

Does it need to be truly accurate?

Depending on what you're trying to accomplish it might be good enough to just graph 10% or 1% of the data volume. (of course, it might also be completely useless, but it all depends on what the visualization is for)

answered May 27 '09 at 16:34



[jplindstrom](#)

757 1 7 11

Check out the Java/Jython based GUESS:
<http://graphexploration.cond.org/>

answered Mar 26 '09 at 22:37



[thestoneage](#)

112 1 9

I don't think you can come remotely close to visualising that in a flat layout.

I've been intrigued by [Hyperbolic Graphs](#), described in [this research paper](#) for some time. Try the software from [SourceForge](#).

Another idea is just graphing the nodes using a [TreeMap](#) as seen at [Panopticode](#).

answered Jan 27 '09 at 13:40



Andy Dent

14k 4 64 92

You could try aiSee: <http://www.aisee.com/manual/unix/56.htm>

answered Jan 6 '09 at 9:22

Johan

1 This link seems to be broken as of 19th May 2015. – [starbeamrainbowlabs](#) May 19 '15 at 7:21

Is there any 'natural' 2d ordering for these nodes (e.g. location) that you could just plot them without drawing the graph?

deleted by [ChrisF](#) ♦ Dec 20 '13 at 13:37

answered Dec 16 '08 at 23:59



OJW

2,014 4 23 40

There's a list of apps here: <http://www.mkbergman.com/?p=414>

Walrus and LGL are two tools supposedly suited for large graphs. However, both seem to require graphs to be input as text files in their own special format, which might be a pain.

answered Nov 2 '08 at 19:07

Rex

Mathematica could very likely handle it, but I have to admit my first reaction was along the lines of the comment that said "take a piece of paper and color it black." Is there no way to reduce the density of the graph?

A possible issue is that you seem to be looking for layout, not just rendering. I have no knowledge about the Big O characteristics of the layouts implemented by various tools, but intuitively I would guess that it might take a *long* time to lay out that much data.

answered Oct 27 '08 at 16:46



Larry OBrien

6,049 1 26 68

4 Mathematica does not handle very large graphs well, not even version 8 with many built-in graph processing functions. The biggest difficulty is that it does not expose the layout algorithm independently of plotting, and its graphics rendering is way too slow to conveniently handle this many edges. – [Szabolcs](#) Jul 29 '11 at 8:21

You might offer a sanitized version of the file to the developers of those tools as a debugging scenario, if all else fails.

answered Oct 27 '08 at 13:03



C. Lawrence Wenham

16.5k 10 50 67

A windows tool that can visualize graphs is [pajek](#), it generates a eps output, however I don't know if it can read your data.

answered Oct 27 '08 at 12:50



Jasper

801 7 13